

Session Assessment - Day 1

1. Research the various types of DNS records and explain the purpose and functionality of each.

Ans: 1) **A Record:** Maps a domain name to an IPv4 address.

2) **AAAA Record:** Maps a domain name to an IPv6 address.

3) **CNAME (Canonical Name):** Creates an alias from one domain name to another domain name.

CNAME points to another domain name, not directly to an IP address.

4) **MX (Mail Exchange):** Specifies which mail servers handle email for a domain.

Directs emails sent to your domain to the correct mail server.

5) **NS Record:** Tells the internet which nameserver oversees your domain's zone.

6) **TXT Record:** Stores text data used to prove domain ownership and secure emails via SPF or DMARC.

7) **SOA (Start of Authority):** Contains important administrative information about a DNS zone.

8) **PTR (Pointer Record):** Performs reverse DNS lookup.
IP --> Domain

9) **SRV (Service Record):** Specifies where a particular network service is available.

2. Describe DNS resolvers, focusing on the roles and mechanisms of cache-only and forwarding resolvers.

Ans: A DNS resolver is a server or service that receives DNS queries from a client and finds the IP address or other DNS information for a domain name.

Cache-only Resolver:

A cache-only resolver stores DNS responses in its cache. When a query is received, it first checks the cache. If the answer is available and its TTL has not expired, it returns the cached answer. Otherwise, it performs a new DNS lookup and stores the result for future requests.

Forwarding Resolver:

A forwarding resolver sends DNS queries to another DNS server called an upstream DNS server when it does not have the answer in its cache. It then receives the response, returns it to the client, and can cache it for future queries.

3. Modify the host file by adding an entry for “google.com [IP]” and analyze the resulting behavior, including the reason for the observed outcome

Ans: The /etc/hosts file can be used to manually map a domain name to an IP address. If google.com is mapped to a specific IP in the hosts file, the system uses that IP instead of performing a normal DNS lookup, because the hosts file is checked before DNS resolution.